

# Emirates Schools Establishment — Maths Diagnostic Performance Analytics (Grades 11 & 12)

Portfolio project demonstrating a repeatable analytics pipeline, data quality controls, cohort segmentation, and KPI reporting.

## Executive KPIs

| timepoint | n  | mean | median | std | pct_ge80 | pct_ge90 | pct_lt70 |
|-----------|----|------|--------|-----|----------|----------|----------|
| Grade 11  | 76 | 78.4 | 80.0   | 7.7 | 55.3     | 11.8     | 7.9      |
| Grade 12  | 76 | 78.9 | 80.0   | 7.4 | 56.6     | 13.2     | 5.3      |

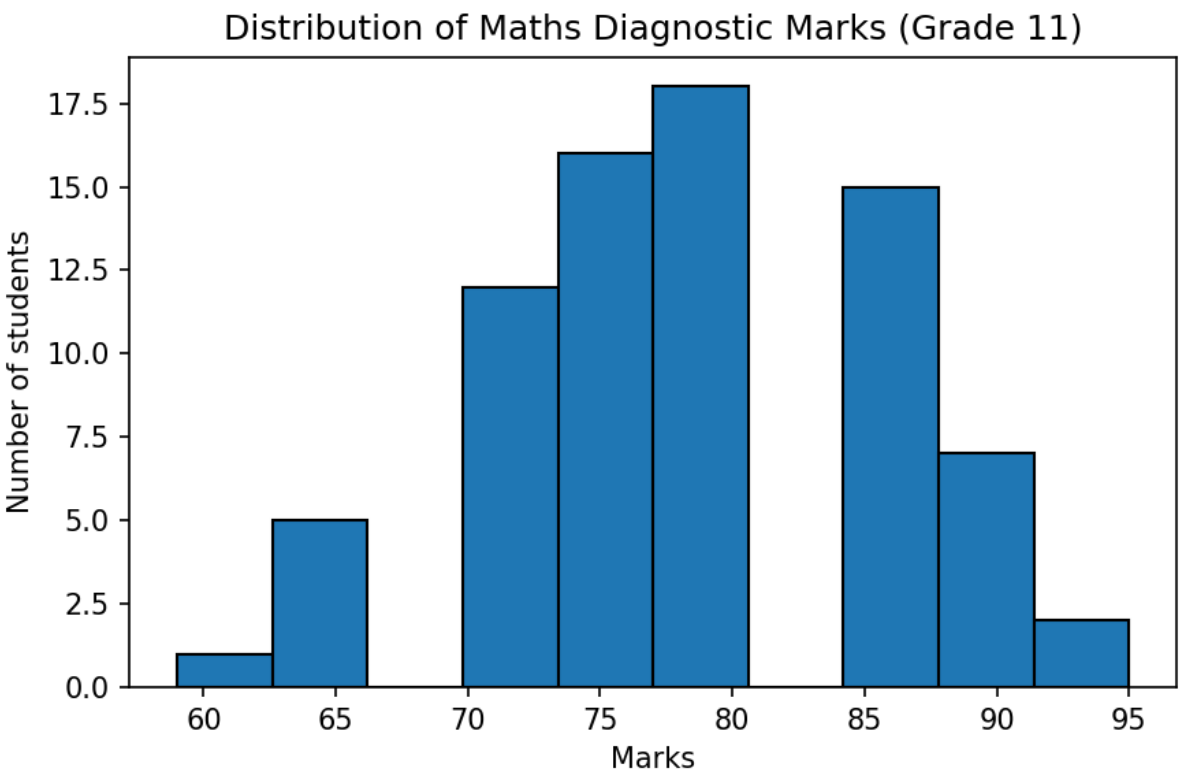
Paired t-test (Grade 11 vs Grade 12):  $p = 0.169$

## KPI Scorecard

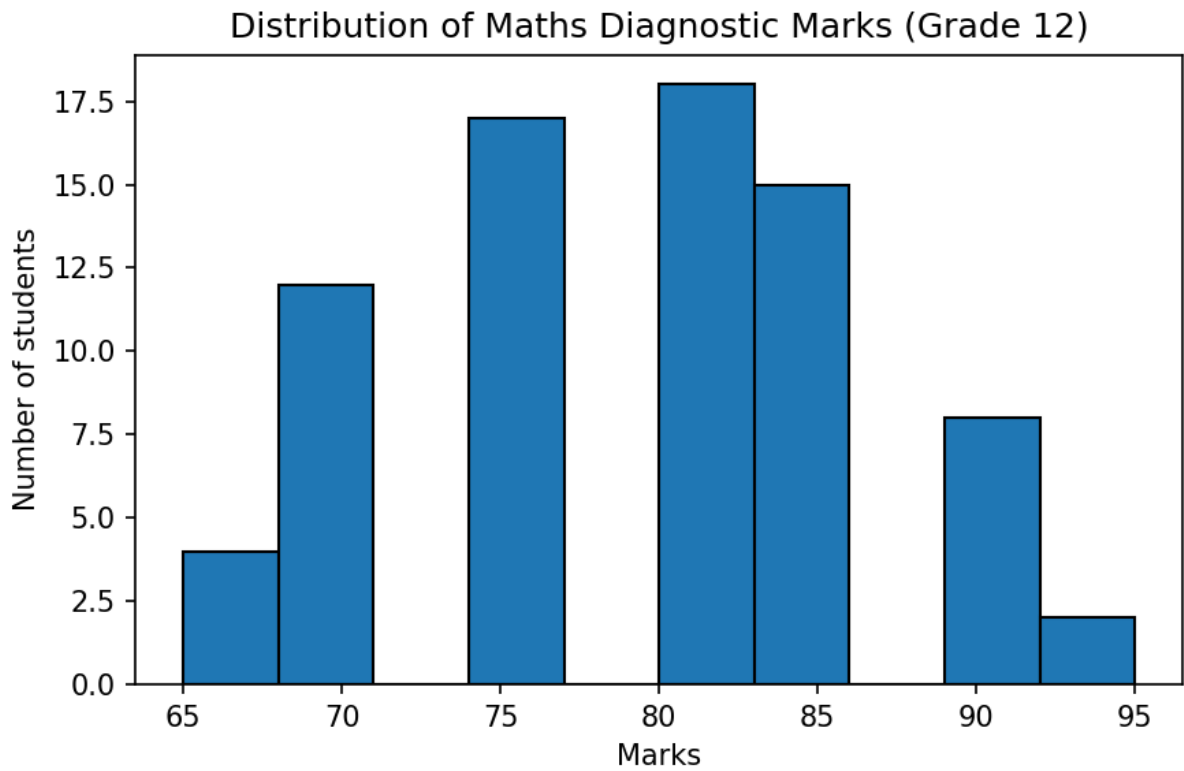
Executive KPI Scorecard: Maths Diagnostic (Grade 11 vs Grade 12)

| Students | Mean (G11) | Mean (G12) | % ≥80 (G12) | % <70 (G12) | Paired t-test p |
|----------|------------|------------|-------------|-------------|-----------------|
| 76       | 78.4       | 78.9       | 56.6%       | 5.3%        | 0.169           |

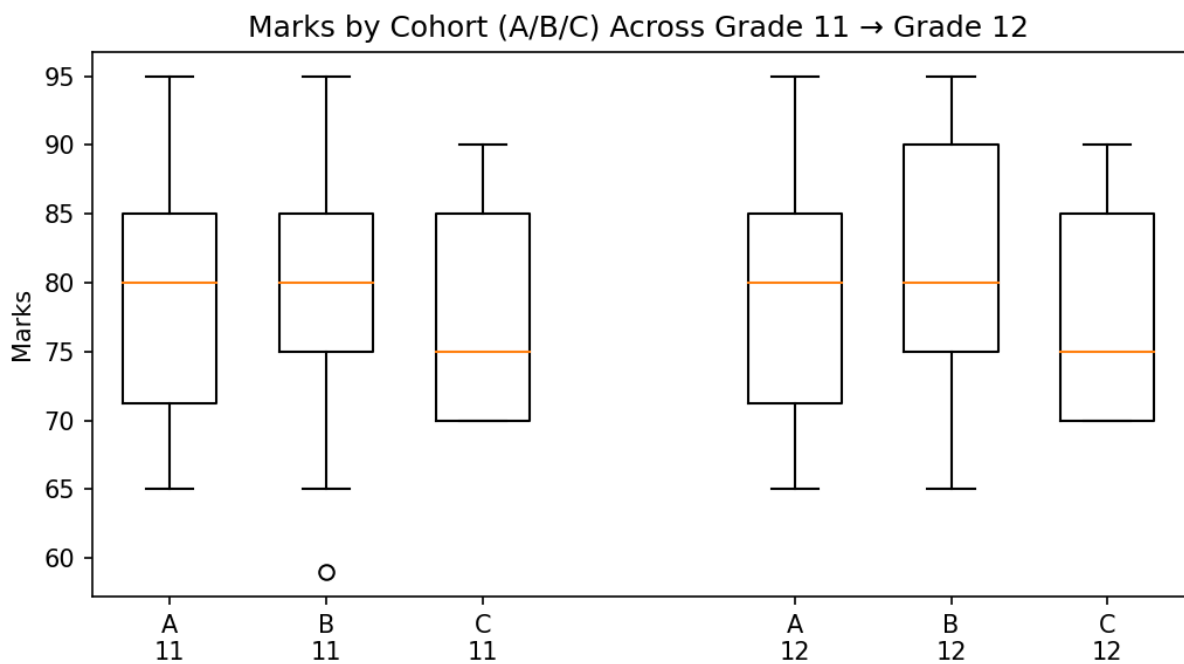
## Distributions — Grade 11



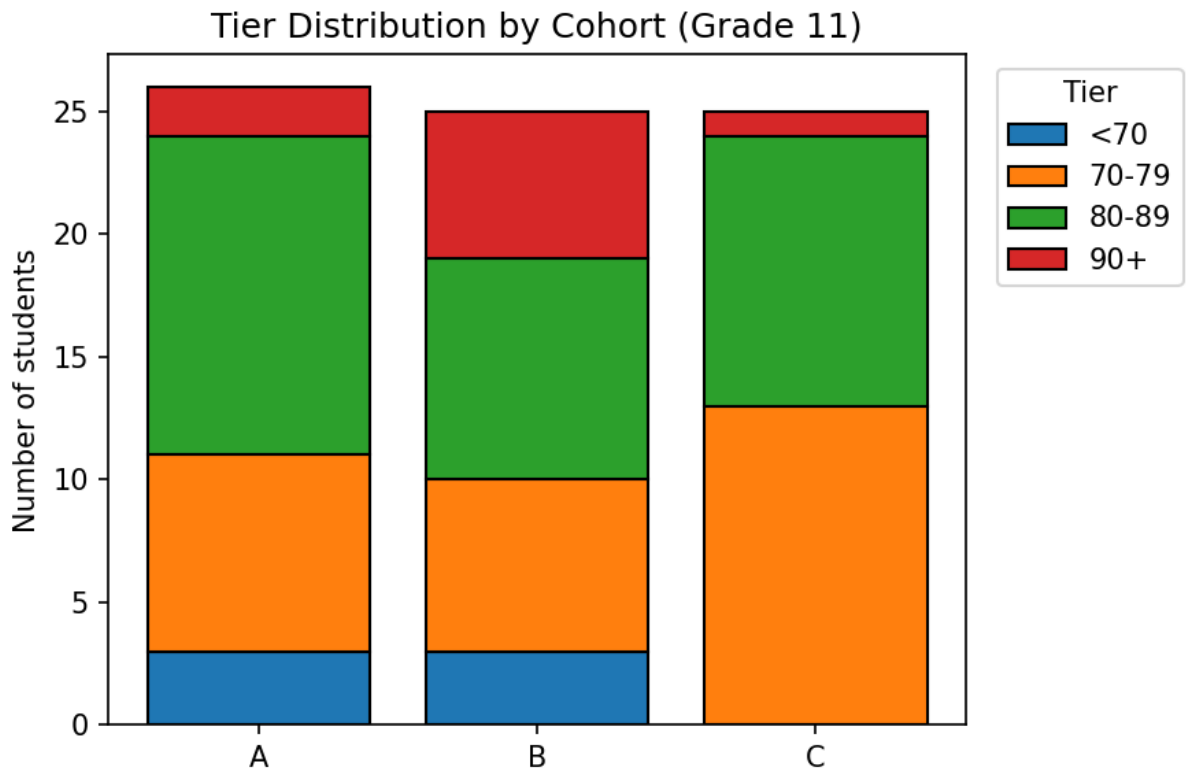
## Distributions — Grade 12



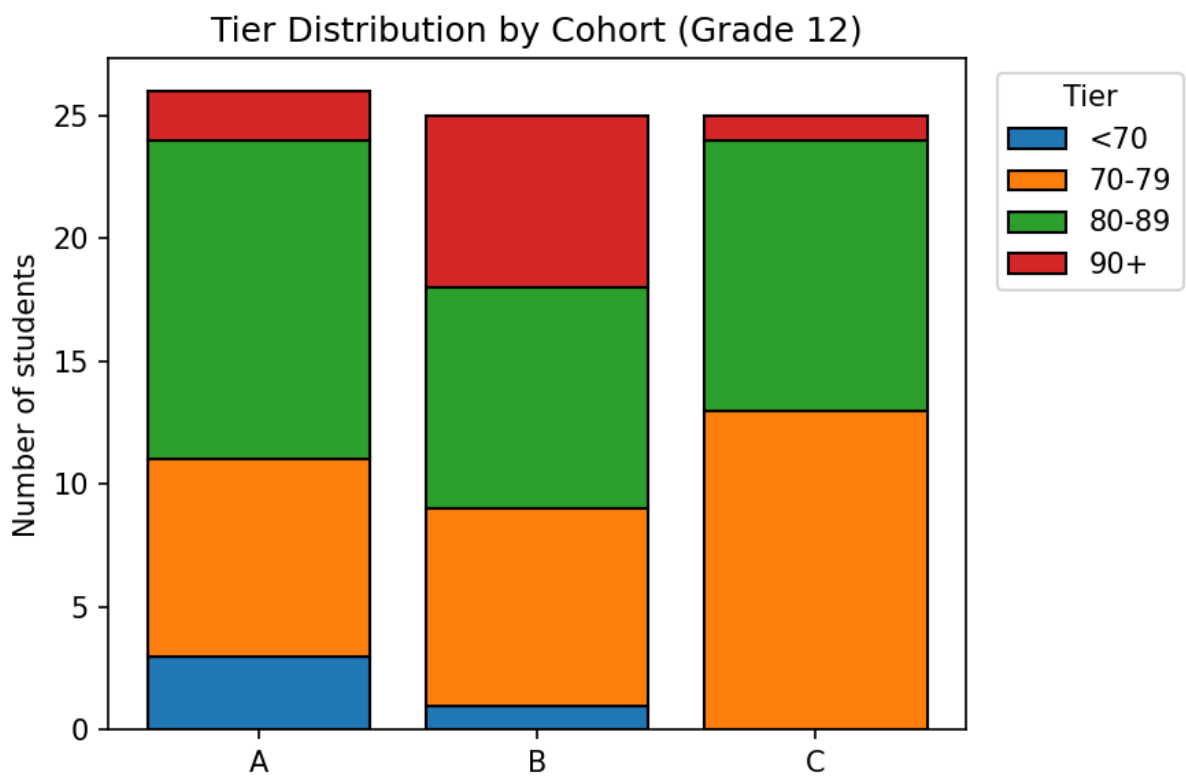
### Cohort boxplot (A/B/C) across Grade 11 → Grade 12



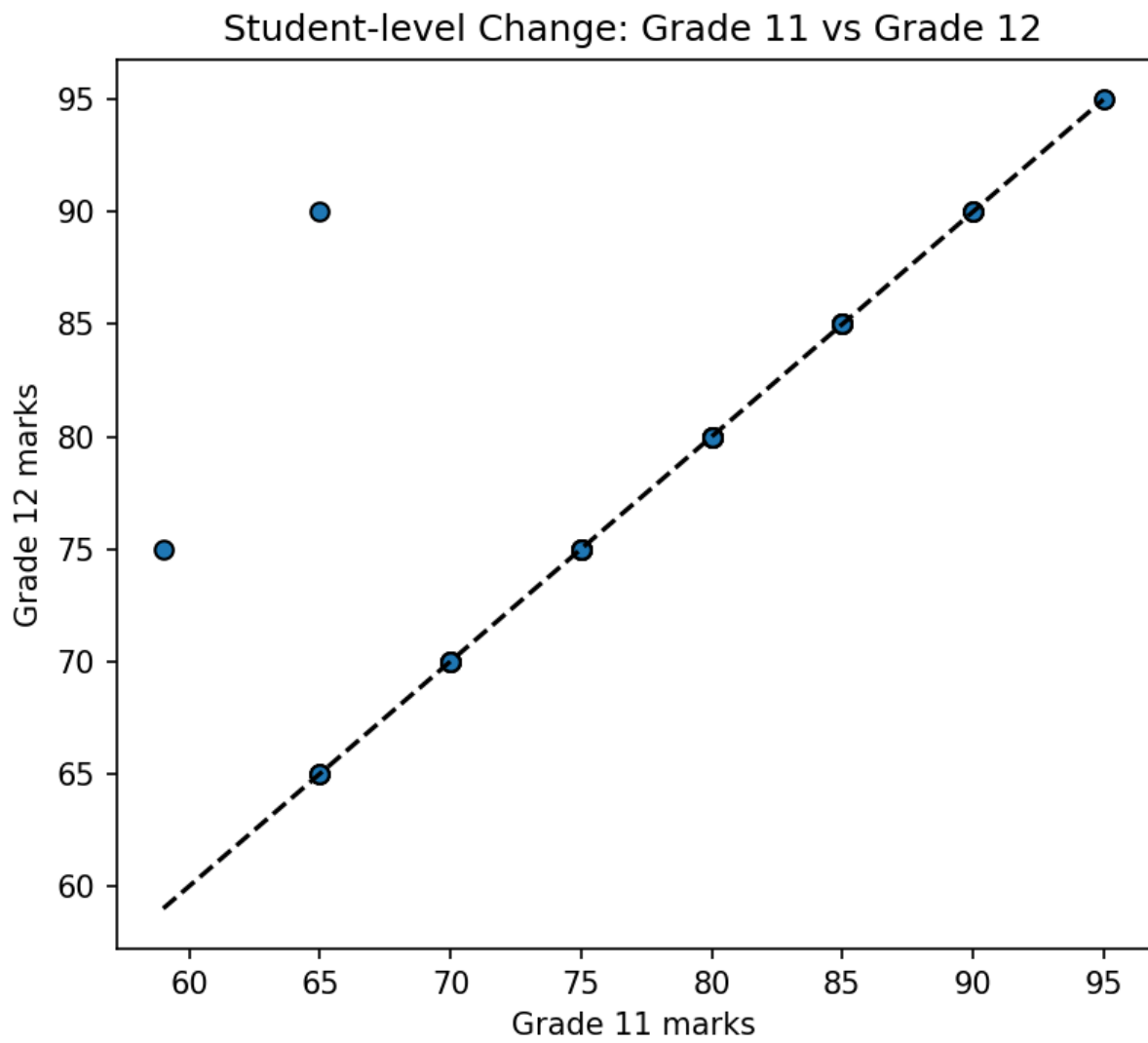
### Tier distribution — Grade 11



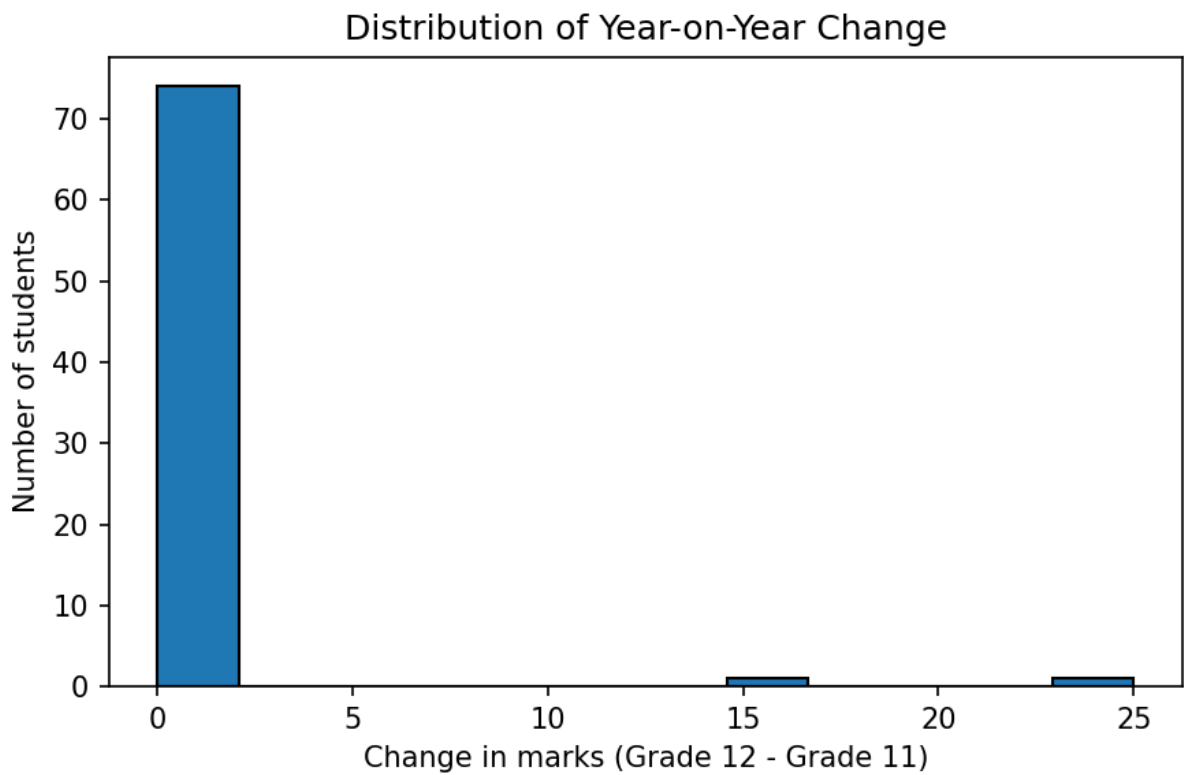
### Tier distribution — Grade 12



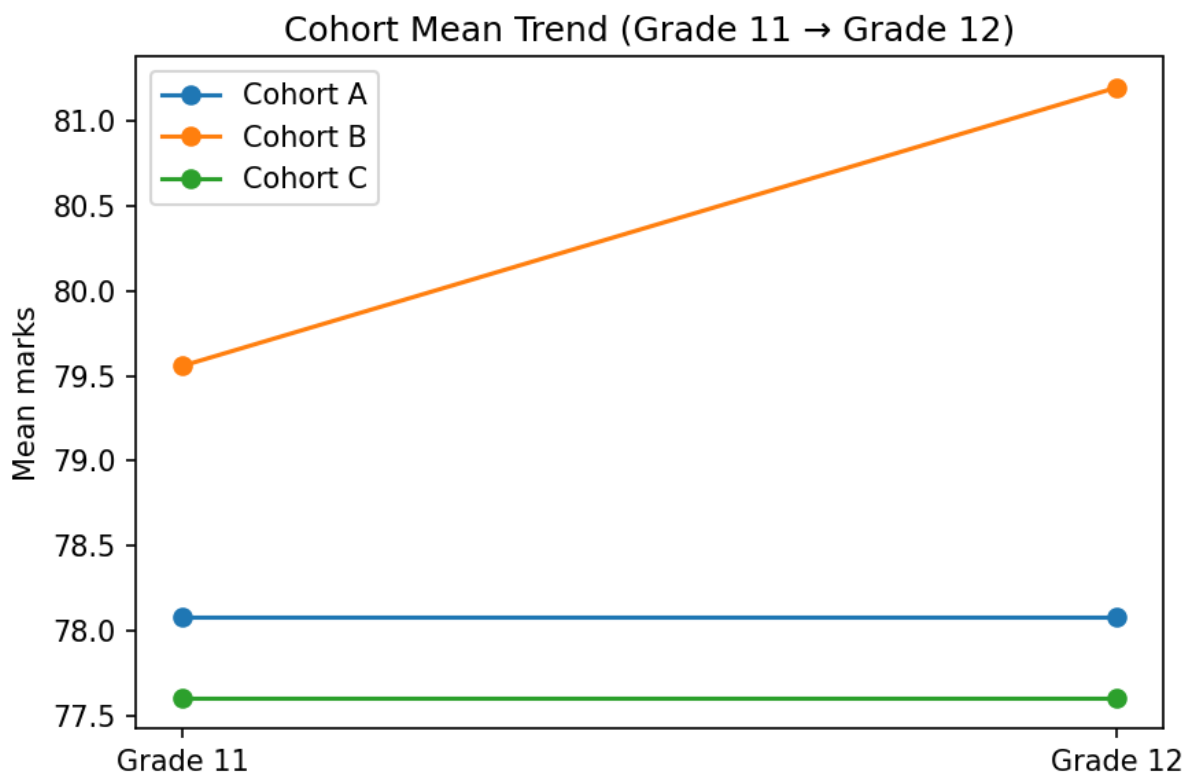
### Student change scatter (Grade 11 vs Grade 12)



**Change distribution (Grade 12 – Grade 11)**



### Cohort mean trend



## **Key Findings (from this sample)**

- Total students: 76.
- Overall mean: Grade 11 = 78.4, Grade 12 = 78.9.
- Benchmark attainment ( $\geq 80$ ) in Grade 12: 56.6%.
- At-risk ( $< 70$ ) in Grade 12: 5.3%.
- Growth distribution: improved=2, declined=0, unchanged=74.

This report is designed as a portfolio artifact and uses anonymized student identifiers suitable for GitHub publishing.